



Kwacha Wize — Android Architecture

Store first, sort second, enrich with AI — 24-hour hackathon build

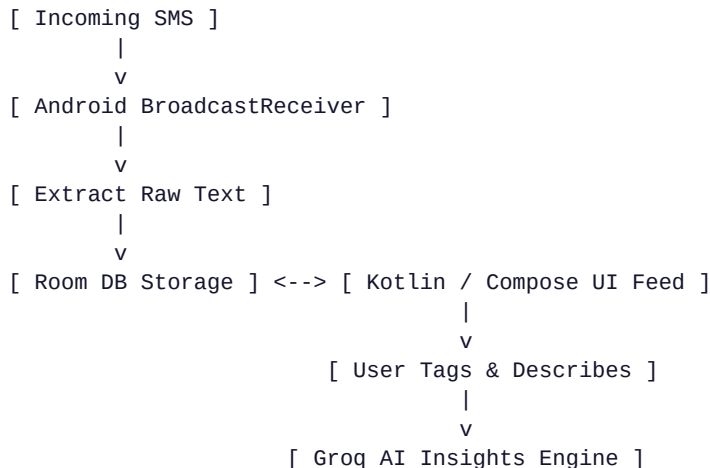
This simplified direction makes complete sense for a fast-paced hackathon setup. Simplifying the app logic to **store first, sort second, and enrich with AI** takes away the headaches of edge-case service detection while keeping the app innovative and practical.

1. Fast, Modern Tech Stack (24-Hour Hackathon Build)

Going native with **Kotlin + Room Database** is a smart move:

- **Kotlin (Jetpack Compose)**: modern declarative UI, no messy XML layouts.
- **Room Database**: Android's official SQLite abstraction. Minutes to set up, offline storage out of the box.
- **Groq API**: ultra-fast, low-cost LLM inference (e.g. llama-3.1-8b-instant) for instant financial insights.

2. Updated Data Flow & System Architecture



The Workflow

- **Silent Ingestion** — a BroadcastReceiver catches any incoming SMS, extracts body, timestamp, and sender, and writes it directly to Room DB as an UNSORTED entry.
- **User Feed (“Review Pending”)** — the user opens KwachaWise to an inbox of unprocessed transactions.
- **Manual Enrichment** — for each entry, the user picks Business or Personal, plus an optional short note (e.g. “Bought stock”, “Lunch”).
- **Groq AI Analysis** — tagged transactions are packaged into a lightweight JSON payload and sent to Groq, which returns weekly summaries, cash-flow trends, and financial advice.

3. Can Regex Handle the Account Balance?

Yes — regex handles balance tracking very reliably. Even without knowing the provider upfront, almost all financial SMS in Malawi follow a distinct pattern near the end of the message:

- **Airtel Money:** “...New balance is MK 45,200.00...” or “New Balance: MWK 45,200”
- **TNM Mpamba:** “...New balance: MK45,200.00...”
- **Bank Alerts:** “...Avail Bal: MWK 45,200.00...”

Quick Regex for Balance Extraction

```
fun parseBalance(smsBody: String): Double? {  
    // Matches "balance", "bal", "avail bal" followed by currency + number  
    val pattern = """"(?i)(?:new\s+balance|avail\s+bal|balance|bal)[:\s]*(?:MWK|MK|K)?\s*([\d, ]+(?:\.\d{2})?)""""  
    val regex = Regex(pattern)  
    val matchResult = regex.find(smsBody)  
    return matchResult?.groupValues?.get(1)?.replace(",", "").toDoubleOrNull()  
}
```

How it updates in Room DB: whenever a new transaction arrives, KwachaWise parses the text with this regex. If a valid balance is found, it updates a CurrentBalance state. The home screen shows this as the user's latest verified wallet/account balance.

4. Kotlin + Room Quick Setup

A. Room Entity — TransactionEntity.kt

```
import androidx.room.Entity  
import androidx.room.PrimaryKey  
  
@Entity(tableName = "transactions")  
data class TransactionEntity(  
    @PrimaryKey(autoGenerate = true) val id: Int = 0,  
    val rawSms: String,  
    val sender: String,  
    val timestamp: Long,  
    val parsedAmount: Double?,  
    val detectedBalance: Double?,  
    var type: String = "UNSORTED", // "BUSINESS", "PERSONAL", or "UNSORTED"  
    var description: String = ""  
)
```

B. Room DAO — TransactionDao.kt

```
import androidx.room.*  
import kotlinx.coroutines.flow.Flow  
  
@Dao  
interface TransactionDao {  
    @Insert(onConflict = OnConflictStrategy.REPLACE)  
    suspend fun insertTransaction(transaction: TransactionEntity)  
  
    @Query("SELECT * FROM transactions ORDER BY timestamp DESC")  
    fun getAllTransactions(): Flow<List<TransactionEntity>>  
    @Query("UPDATE transactions SET type = :type, description = :description WHERE id = :id")
```

```
suspend fun updateTransactionTag(id: Int, type: String, description: String)
}
```

5. Groq API Integration (JSON Payload)

When the user requests financial insights, filter for tagged transactions (BUSINESS or PERSONAL) and post a single structured request to Groq:

```
{
  "model": "llama-3.1-8b-instant",
  "messages": [
    {
      "role": "system",
      "content": "You are KwachaWise, an AI financial coach for Malawian
        micro-entrepreneurs. Analyze the categorized transactions and give
        3 short, actionable financial recommendations."
    },
    {
      "role": "user",
      "content": "[{\"type\":\"BUSINESS\",\"amount\":15000,\"desc\":\"Bought inventory\"},
        {\"type\":\"PERSONAL\",\"amount\":2000,\"desc\":\"Airtime\"}]"
    }
  ]
}
```

This stack gives you a fast Kotlin native app, offline persistence via Room, reliable balance tracking via regex, and low-cost AI insights powered by Groq.